

Welcome!!



Computer Systems & Networks

Dr Frank Walsh, Caroline Cahill, Mujahid Tabassum, SETU

FRIDAY 16TH JAN 2026

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Computer Systems & Networks

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Caroline Cahill
Caroline Cahill



Frank Walsh(Lecturer) ○
Frank Walsh



Mujahid Tabassum

Computer Systems & Networks

10 Credit

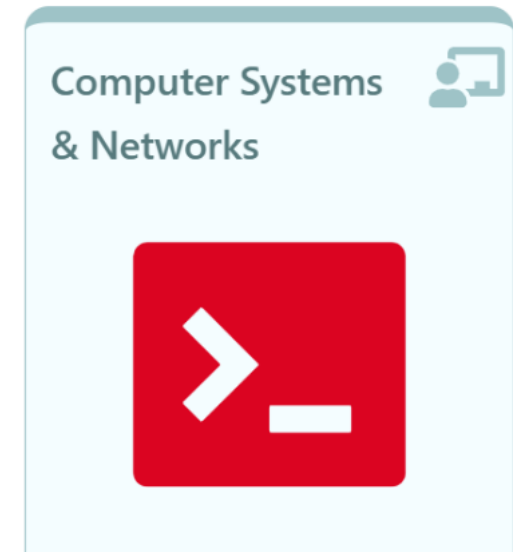
Module delivery split between

☐ Caroline for Computer Systems

and

☐ Frank & Mujahid for Networks

Timetabled for TWO sessions per week



2025

Semester 2

S M T W T F S Modules

January

1	18	19	20	21	22	23	24	comp sys & database
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2	25	26	27	28	29	30	31	comp sys & database
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February

3	1	2	3	4	5	6	7	comp sys & database
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4	8	9	10	11	12	13	14	
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<i>reading-week</i>	15	16	17	18	19	20	21	comp sys & database
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5	22	23	24	25	26	27	28	comp sys & database
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March

6	1	2	3	4	5	6	7	comp sys & database
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<i>reading-week</i>	8	9	10	11	12	13	14	comp sys & database
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7	15	16	17	18	19	20	21	
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8	22	23	24	25	26	27	28	comp sys & database
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April

<i>EASTER</i>	29	30	31	1	2	3	4	comp sys & database
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<i>EASTER</i>	5	6	7	8	9	10	11	
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9	12	13	14	15	16	17	18	comp sys & database
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10	19	20	21	22	23	24	25	comp sys & database
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11	26	27	28	29	30	1	2	comp sys & database
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May

12	3	4	5	6	7	8	9	comp sys & database
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<i>reading-week</i>	10	11	12	13	14	15	16	
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<i>reading-week</i>	17	18	19	20	21	22	23	
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Computer Systems & Networks



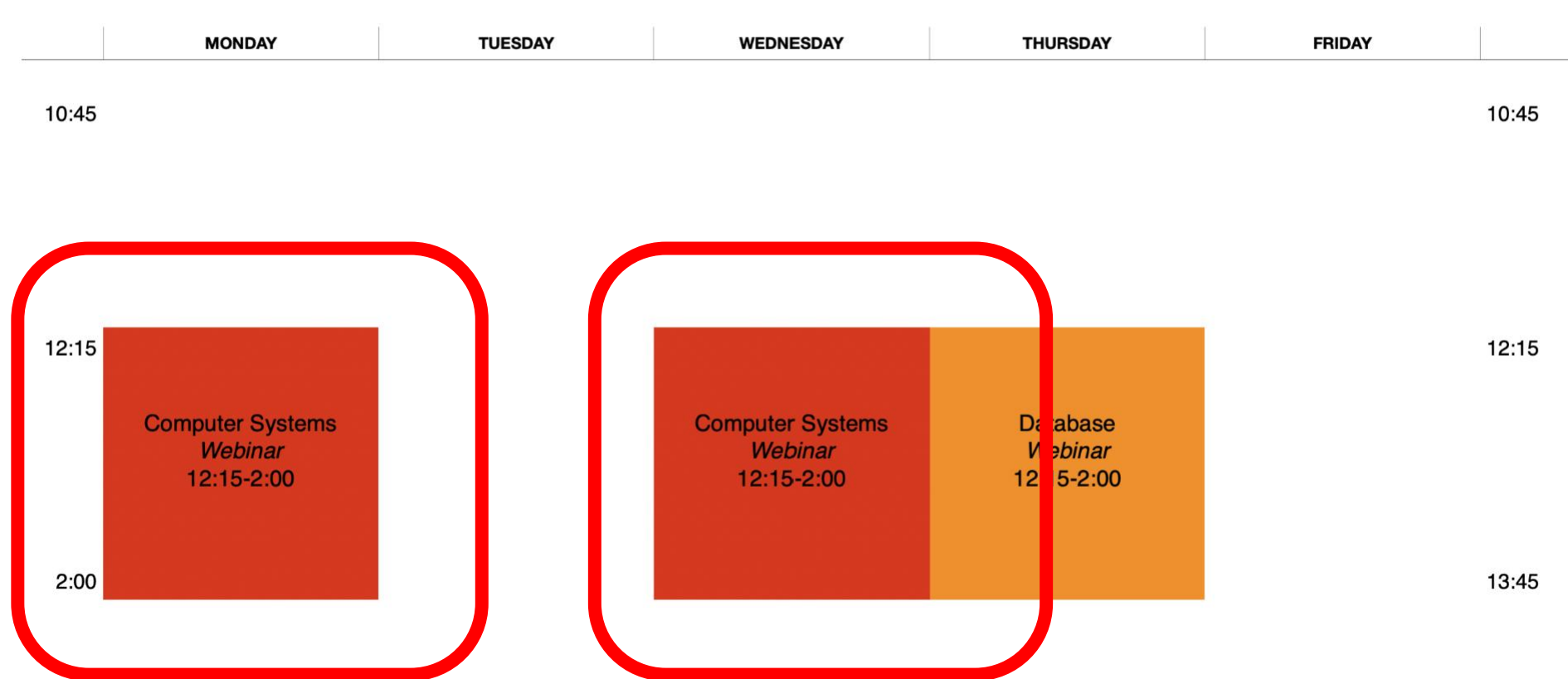
logic · computer
organisation · os · networks
· interfaces · sensors

Database



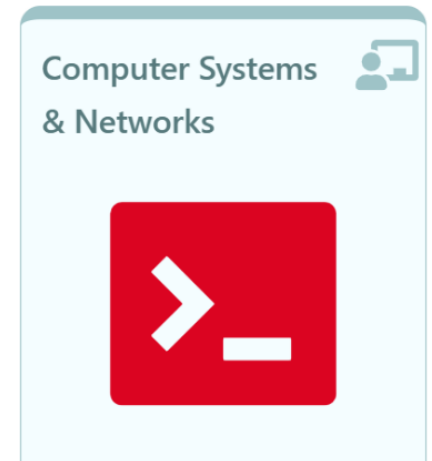
entities · tables · rows · sql
· er · nosql

Weekly Webinar Schedule

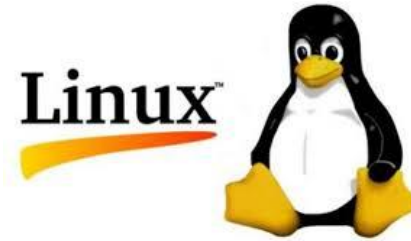


Module Overview(Tentative):

- Number bases used in Computer Science
- Boolean logic
- Computer system architecture
- Operating systems: Components, services, and utilities
- Memory and file management
- Scripting and shell programming
- Virtualisation and hypervisors
- Internet protocol suite
- Physical/network addressing
- Transport layer protocols
- Application layer protocols
- Wireless network protocols: LAN and PAN



Development Tech (tentative)



```
#!/bin/bash
# This is a basic bash script.
a=Hello
b="Good Morning"
c=16

echo $a
echo $b
echo $c
```

Cisco Packet Tracer



Build Your Skills With Cisco

Pursue real career paths through instructor-led courses taught by experts and free, online courses backed by Cisco's expertise.



Cisco NetAcad

YOU WILL COMPLETE UP
TO AND INCLUDING
MODULE 11

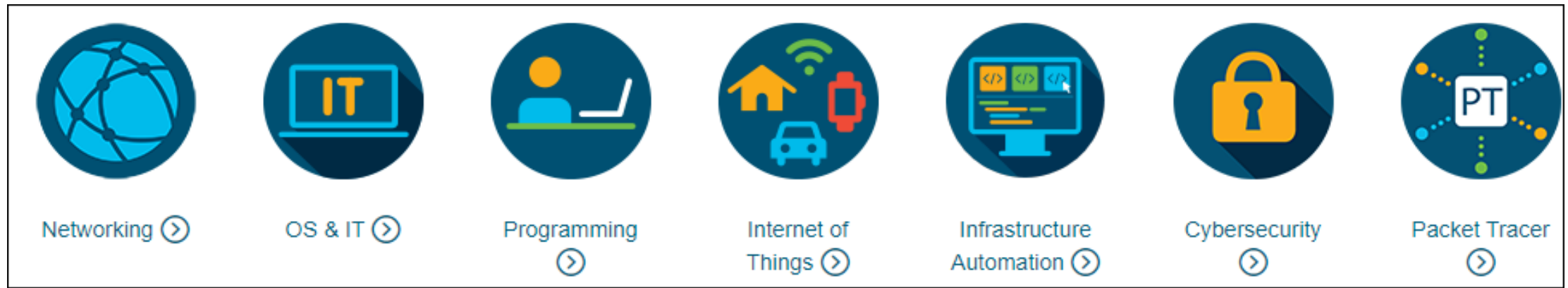
What is Cisco Network Academy?

Empowering educators with a next-gen teaching experience

Cisco Networking Academy is a global IT and cybersecurity education program that provides:

- best-in-class courses,
- tools, and
- resources

to help people from all backgrounds participate in the digital economy.





Global Impact Report

Educating the learners who connect the world

Netacad Students Around the World:

In-Kind Contributions



\$5.25B USD

since inception, in curriculum, tools, resources, and support to students, schools and instructors



95%

students who took Cisco certification-aligned courses obtained a job and/or educational opportunity based on surveys since fiscal 2005 through fiscal 2023

Students



3,735,000

current annual students



20,496,000

total students since inception



27%

are female since inception
based on those who identified gender

Instructors



29,700

active instructors



22%

are female
based on those who identified gender

Cisco Academies



11,700

institutions and organizations offering
Networking Academy courses

2024

- 4.7 million current annual learners
- 12,100 institutions and organizations offering courses in **191** countries.

Networks and Cybersecurity Jobs in Ireland

in networks Ireland Se

Jobs ▾ Date posted ▾ Experience level ▾ Com

networks in Ireland 81 results Set alert ☐

-  **Network Architect** ... x
E-Frontiers
Dublin, County Dublin, Ireland (On-site)
Promoted • Be an early applicant
-  **Aruba Network Engineer** ... x
Linnk Group
Dublin, County Dublin, Ireland (On-site)
Actively reviewing applicants
Promoted • Easy Apply
-  **IT Manager** ... x
Auxilia Group Recruitment
Dublin, County Dublin, Ireland (Hybrid)
Actively reviewing applicants
Promoted • Easy Apply
-  **Core Network Engineering Manager** ... x
Cubic³
Dublin, County Dublin, Ireland (Hybrid)
Promoted • Easy Apply
-  **Network Architect** ... x
Fruition Group Ireland
Dublin, County Dublin, Ireland (Hybrid)
€70/hr - €85/hr

Jan 07,
2026

-  **Senior Network Engineer** ✓
HCLTech
Dublin, County Dublin, Ireland (On-site)
Actively reviewing applicants
Promoted • Easy Apply
-  **Network Implementation Engineer, Core Design** ✓
Google
Dublin, County Dublin, Ireland (On-site)
5 connections work here
Promoted
-  **Network Security Engineer** ✓
TPx
Ireland (Remote)
Actively reviewing applicants
Promoted • Easy Apply
-  **Defence Forces Civilian Cyber Responder**
Department of Defence Ireland
Dublin, County Dublin, Ireland (On-site)
Promoted
-  **Virtualization Engineer** ✓
Millennium
Dublin, County Dublin, Ireland
Promoted
-  **Senior Network Engineer** ✓
Nicoll Curtin
County Dublin, Ireland (On-site)
Actively reviewing applicants

Jobs Market for Networking and Cyber Security Graduates

By 2025, automation and a new division of labor
between humans and machines will disrupt

85 million jobs

globally in medium and large businesses across 15
industries and 26 economies.

97 million new jobs

will be created by 2025 due to AI.

SOURCE: [Future of Jobs Report](#), World Economic Forum

Cybersecurity Jobs. PHOTO: Cybercrime Magazine.

Cybersecurity Jobs Report: 3.5 Million Openings In 2025

Career Path

Skills-to-Jobs Curriculum

A comprehensive portfolio



Networking

Gain hands-on, relevant networking skills



Programmable Infrastructure

Learn programming, infrastructure automation, and Internet of Things



Cybersecurity

Learn to secure and defend networks



Operating Systems and IT

Essential skills for the digital world



Programming

Learn to code in languages like Python, C, or C++



Practice

Interactive tools and experiences build real skills, not just knowledge

Cisco Certifications



BEGINNER

allied Networking
CISCO Academy

📖 Course | Self-paced

Introduction to IoT and Digital Transformation

Billions of devices connect to the network every day. Learn how IoT is digitally transforming the world and opening up exciting new jobs.

🕒 6 Hours 🔒 Free



INTERMEDIATE

allied Networking
CISCO Academy

📖 Course | Instructor-led

Networking Essentials

Networks keep the digital world connected. Learn how networks work and gain employable skills to start your tech career.

🕒 70 Hours



BEGINNER

allied Networking
CISCO Academy

📖 Course | Self-paced, Instructor-led

Networking Basics

Start learning the basics of computer networking and discover how networks operate.

🕒 22 Hours 🔒 Free

Career Paths | [View all](#)



BEGINNER

allied Networking
CISCO Academy

🎓 Career Path

Network Technician

Catalog > Linux Essentials

South East Technological University Course

Linux Essentials

Welcome to SETU Computer Systems & Networks mod

SCHEDULE
Aug 01, 2025 - Jul 28, 2026

Get Started Invite Learners

Overview Find An Academy Resources



Instructor-led Version
Learn from an educator to enrich and custom

Linux, an open source operating system, has a global r much more. Whether you want to become a system ac learning Linux is foundational.

This course quickly builds your Linux knowledge and gi prepares you for the [Linux Professional Institute Linux](#) your new skills to employers. The Linux Essentials certi

This course is developed in collaboration with [Network](#)

Welcome to NDG Linux Essentials!	Due Date	11 Jan 2026 11:59 PM	...	>
Help & Resources	Due Date	11 Jan 2026 11:59 PM	...	>
Before You Get Started	Due Date	11 Jan 2026 11:59 PM	...	>
Module 1 - Introduction to Linux	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 1	Due Date	11 Jan 2026 11:59 PM	...	>
Module 2 - Operating Systems	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 2	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 02 Exam	Due Date	11 Jan 2026 11:59 PM	...	>
Module 3 - Working in Linux	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 3	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 03 Exam	Due Date	11 Jan 2026 11:59 PM	...	>
Module 4 - Open Source Software and Licensing	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 4	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 04 Exam	Due Date	11 Jan 2026 11:59 PM	...	>
Module 5 - Command Line Skills	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 5	Due Date	11 Jan 2026 11:59 PM	...	>
Lab 05	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 05 Exam	Due Date	11 Jan 2026 11:59 PM	...	>
Module 6 - Getting Help	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 6	Due Date	11 Jan 2026 11:59 PM	...	>
Lab 06	Due Date	11 Jan 2026 11:59 PM	...	>
Chapter 06 Exam	Due Date	11 Jan 2026 11:59 PM	...	>



18 LABS INSTRUCTOR-LED

Using Linux

- ❖ You can use [WSL](#) (windows subsystem for linux)

- ❖ If using a Mac, the standard Mac terminal will be perfect

NOTE: If you use the mac terminal then you are working and making changes to your actual OS

You could use any emulator/any virtualiser software & then install Linux on top of this

(= keeps your Linux completely separate from your work system)

- ❖ You could install [VirtualBox](#) & then install **Lubuntu** onto this

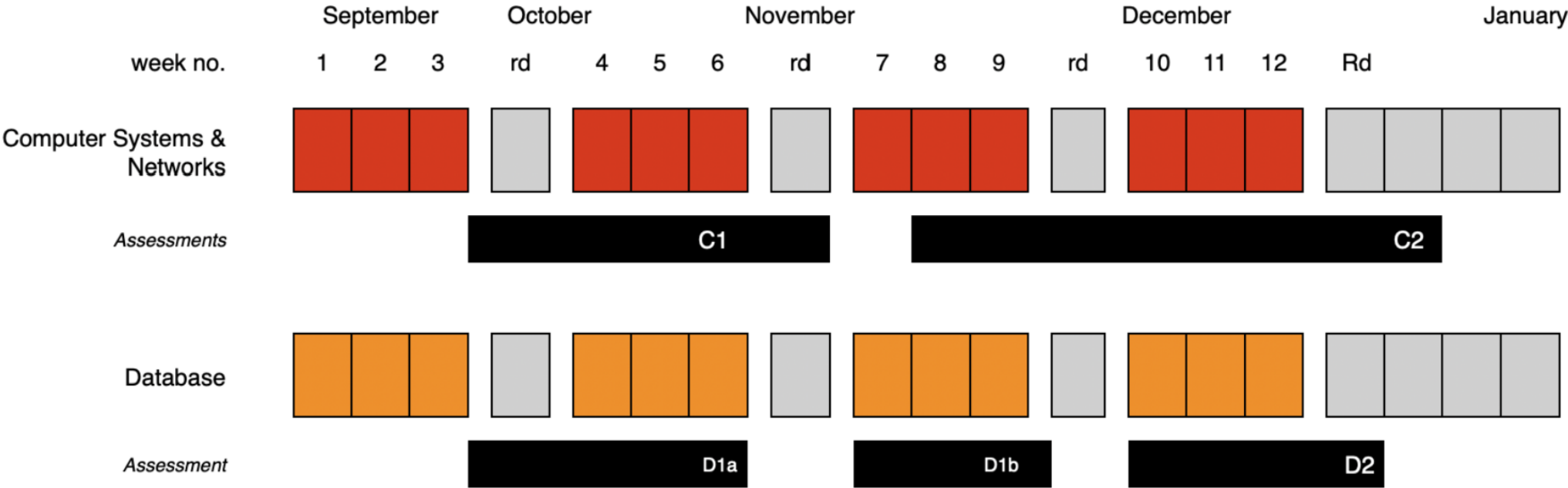
VirtualBox also won't install on a M1/M2 chip as they removed support for it

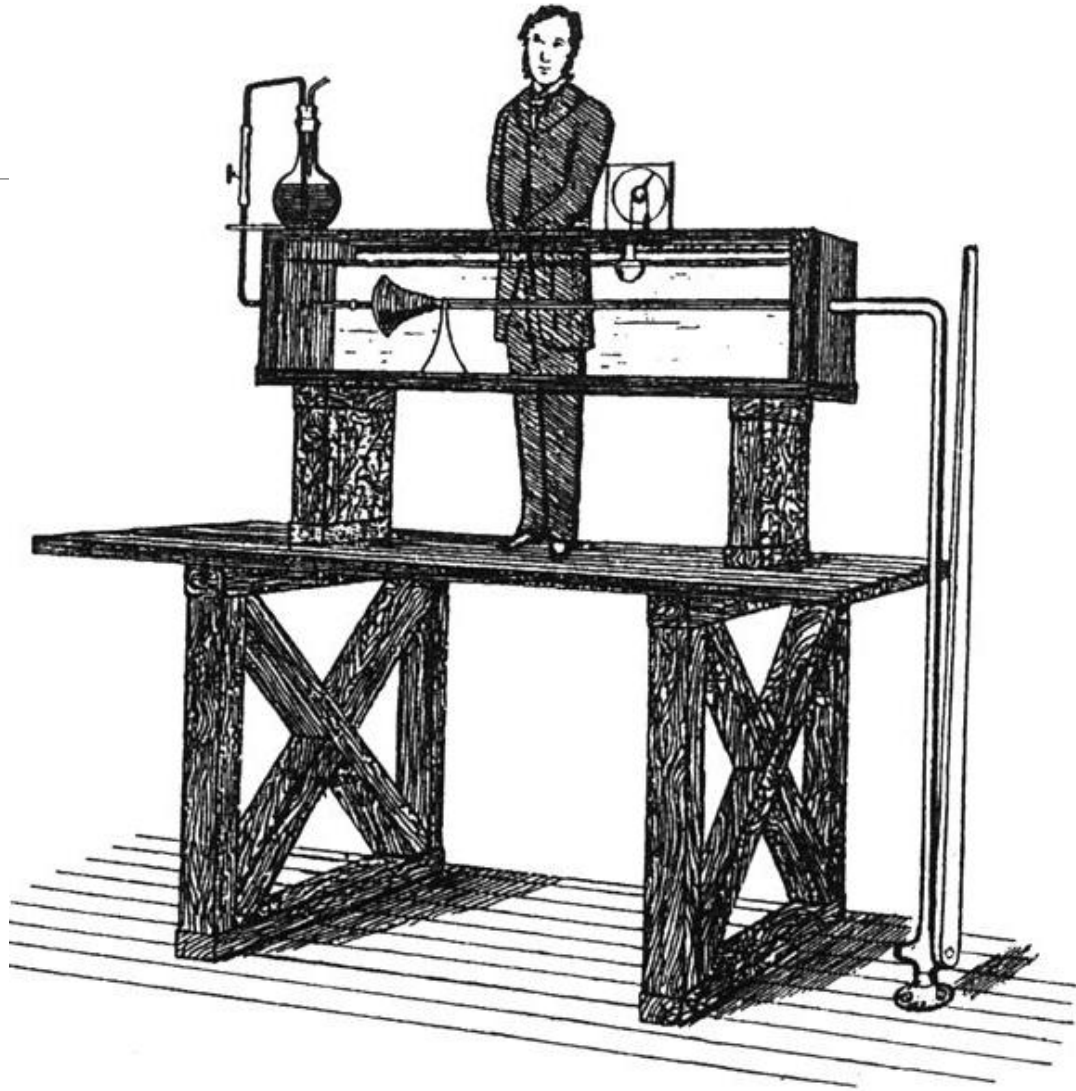
- ❖ You can use a UTM virtualiser (was created for macOS & only for Apple platforms)

- ❖ You can install Lubuntu on a Raspberry Pi (the SD card will hold everything)

Many of screenshots in the notes will be taken off my own Lubuntu that I've installed in VirtualBox

Semester 2 Assessment Schedule





Ethos

Focus on practical skills

Good idea to stay current with module:

- Some lectures will require reading/viewing of talks before.

Experiment/build something interesting:

- You will have the opportunity to propose your own assignment!

Just please have your linux
system of choice ready to
start for classes 19th Jan



It's your choice

Which
will you
use?

Allow yourself
to be a beginner.
No one starts off
being excellent.